Quaternionic characters in a vertex permutation representation

Human Compression project: recovery of L024

We recover the example and infinite-family conclusion of Cameron, Herman and Leemans [1]. The example answers Monson's existence question and its polytopal variant; these answers belong to that source. Our account fixes explicit matrices, identifies the subgroup fusion without a character-table search, and uses a common character bound to shorten the positivity verification. It also separates an erroneous small-dimensional case in the printed infinite-family statement.

1 Fixed vectors and a dihedral stabilizer

Let G be finite and $H \le G$. For an irreducible complex character χ, Frobenius reciprocity and the averaging projection give

$$[1_H^G, \chi] = \dim V_\chi^H = \frac{1}{|H|} \sum_{h \in H} \chi(h). \tag{1}$$

Indeed $|H|^{-1} \sum_{h \in H} \rho(h)$ is an idempotent with image the fixed subspace, and its trace is the right side. The permutation module on G/H has that same multiplicity by reciprocity. For a vertex stabilizer in a regular polytope, the nontrivial constituents are precisely the polytopal characters in the terminology of [1]. The trivial constituent is removed in the deleted permutation module.

Lemma 1. *Suppose H is dihedral of order 2ℓ, where ℓ is an odd prime. Assume all its reflections lie in a G-class t^G and all its nonidentity rotations lie in a G-class u^G. Then*

$$[1_H^G, \chi] = \frac{\chi(1) + \ell\chi(t) + (\ell - 1)\chi(u)}{2\ell}.$$

In particular this multiplicity is positive whenever

$$\chi(1) > \ell\sqrt{|C_G(t)|} + (\ell - 1)\sqrt{|C_G(u)|}.$$

Proof. There are ℓ reflections and $\ell - 1$ nonidentity rotations, so the formula follows from (1). Column orthogonality gives $\sum_{\psi \in \mathrm{Irr}(G)} |\psi(x)|^2 = |C_G(x)|$. Consequently $|\chi(x)| \le \sqrt{|C_G(x)|}$. Applying the triangle inequality to the numerator yields the sufficient condition. The average is a nonnegative integer by (1). □

We also retain the Frobenius–Schur theorem: the indicator

$$\nu_2(\chi) = |G|^{-1} \sum_{x \in G} \chi(x^2)$$

is -1 precisely for a real-valued irreducible character of real Schur index two. These are standard character-theoretic inputs, not new theorems of this work.

2 A fixed marking of $\mathrm{PSU}(5,3)$

Let $E = \mathbb{F}_3[z]/(z^2 - z - 1)$, let J be the 5 by 5 antidiagonal identity, and write $\bar{x} = x^3$. We use row vectors and the unitary condition $sJ\bar{s}^{,t} = J$. Define

$$s_0 = \mathrm{diag}(-1,1,1,1,-1), \qquad s_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix},$$

and

$$s_1 = \begin{pmatrix} z^7 & z^3 & z^7 & 0 & z^4 \\ z^5 & z^6 & 0 & z^4 & 0 \\ z^3 & z^4 & z^4 & 0 & z^5 \\ z^5 & z^4 & z^4 & z^2 & z \\ 0 & z^7 & z & z^7 & z^5 \end{pmatrix}.$$

The center of $\mathrm{SU}(5,3)$ has order $\gcd(5,4) = 1$, so this group equals $G = \mathrm{PSU}(5,3)$. Its standard unitary order formula gives $|G| = 258190571520$.

Proposition 2. *The ordered triple (s_0, s_1, s_2) is a string C-group marking of G of type $\{5,5\}$. Its vertex stabilizer $H = \langle s_1, s_2 \rangle$ is dihedral of order ten.*

Proof. Direct arithmetic in E verifies that all three matrices have determinant one, preserve J, and are involutions. It gives

$$|s_0 s_1| = |s_1 s_2| = 5, \qquad |s_0 s_2| = 2.$$

For the finite generation step we use the exact attached certificate, `verify_schur_example.g`. It enumerates the projective isotropic points by normalizing the first nonzero coordinate to one and testing $\sum_i v_i v_{6-i}^3 = 0$. There are 2440 points. The three displayed matrices act on this literal set. A stabilizer chain for their permutation group has successive orbit lengths

$$2440, \quad 2187, \quad 2, \quad 224, \quad 36, \quad 3.$$

Their product is 258190571520. Thus the subgroup generated by the matrices has an image of order $|G|$, and containment in G proves equality. The verification needs no search for the generator triple or downloaded group representation. The permutation stabilizer-chain option `random=1000` requires guaranteed correctness of the verification, even though the internal construction may use random choices; the certificate asserts this setting. The chain itself is recomputed exactly from the fixed matrices; the displayed product alone would not establish generation.

We now prove the intersection property from the product orders and full generation. Two involutions whose product has order five generate a dihedral group of order ten. Thus the adjacent parabolics $P_{01} = \langle s_0, s_1 \rangle$ and $P_{12} = \langle s_1, s_2 \rangle$ are both dihedral of order ten, whereas $P_{02} = \langle s_0, s_2 \rangle$ is a Klein four-group. A subgroup of a dihedral group of order ten that contains a reflection has order two or ten. Since $P_{01} \cap P_{12}$ contains s_1, a larger intersection would make the two parabolics equal and put all three generators in a group of order ten. Full generation excludes this, so their intersection is $\langle s_1 \rangle$. The intersection of P_{02} with either adjacent parabolic has order dividing $\gcd(4,10) = 2$; its common displayed generator therefore generates the whole intersection. A singleton generator cannot belong to its opposite parabolic, since that too would put all generators in a proper parabolic. Distinct singleton subgroups meet trivially, and all containment, empty-set and whole-group cases are immediate. Consequently for every $I, K \subseteq \{0,1,2\}$,

$$\langle s_i : i \in I \rangle \cap \langle s_i : i \in K \rangle = \langle s_i : i \in I \cap K \rangle.$$

The certificate also checks all 64 pairs as corroboration; that part of the computation is not a premise of this intersection argument. The generation computation remains a finite premise, not a Lean formalization. □

The source's two centralizer choices can also be identified directly. An involution of G has a nondegenerate negative eigenspace of dimension two or four. The corresponding class sizes are 444690 and 4941, with centralizer orders 580608 and 52254720. Our three matrices have negative eigenspace dimension two. For s_0, write its orthogonal decomposition as $A \perp B$, with $\dim A = 2$, $\dim B = 3$. Then

$$C_G(s_0) = \{(a, b) \in \mathrm{GU}(A) \times \mathrm{GU}(B) : \det(a) \det(b) = 1\}.$$

The negative eigenspace dimensions (k, l) of an involution in this centralizer must have positive even sum. The five possibilities are $(0, 2), (1, 1), (1, 3), (2, 0), (2, 2)$. Their class sizes are $63, 378, 6, 1, 63$. To see this, unitary groups are transitive on nondegenerate subspaces of each dimension, and the numbers of nondegenerate lines in dimensions two and three over E are

$$\frac{|\mathrm{GU}(2,3)|}{|\mathrm{GU}(1,3)|^2} = 6, \qquad \frac{|\mathrm{GU}(3,3)|}{|\mathrm{GU}(1,3)|\,|\mathrm{GU}(2,3)|} = 63.$$

The determinant-one constraint does not split these orbits: the centralizer of either eigenspace decomposition has a determinant surjective unitary factor, which corrects any conjugator's determinant. The matrix s_2 has $(k, l) = (1, 1)$, so it lies in the prescribed largest class of size 378.

3 The character calculation

We retain the ordinary character table of G distributed by the GAP Character Table Library. It has 150 irreducibles, the two involution classes above, and exactly one class of elements of order five, whose centralizer has order 80. Call the larger involution class $2B$ and the order-five class $5A$. All five reflections of H are conjugate within H and hence lie in $2B$, while its four nonidentity rotations lie in $5A$. This determines the fusion needed for induction without choosing one among possible table fusions. Lemma 1 gives

$$m_\chi = \frac{\chi(1) + 5\chi(2B) + 4\chi(5A)}{10}. \tag{2}$$

Since $\sqrt{580608} < 762$ and $\sqrt{80} < 9$, every irreducible of degree greater than 3846 has positive multiplicity. The retained table has precisely 24 characters below this threshold, and the smallest remaining degree is 4270. For those 24 characters all needed data are displayed below; equal rows are grouped in the first column.

Number	$\chi(1)$	$\chi(2B)$	$\chi(5A)$	m_χ
1	1	1	1	1
1	60	4	0	8
1	61	5	1	9
2	61	9	1	11
1	549	-11	-1	49
1	610	-6	0	58
2	610	22	0	72
1	1220	-12	0	116
2	1220	12	0	128
1	1281	49	1	153
2	1281	21	1	139
1	1830	38	0	202
2	1830	10	0	188
1	1890	42	0	210
2	2440	24	0	256
1	3660	20	0	376
2	3660	-36	0	348

Equation (2) gives the last column. The remaining multiplicities exceed $(4270 - 3846)/10 > 42$. Hence every nontrivial irreducible is polytopal, and the least such multiplicity is eight, attained at degree sixty.

The certificate computes the indicators directly from the class sizes and the square power map, using their defining sum. Exactly five are -1. Their degrees and multiplicities are

$\chi(1)$	60	3660	7320	10980	14580
m_χ	8	376	752	1072	1440.

The character table and its power map are borrowed data. Their construction is not reproved here. The fixed-matrix generation certificate and written intersection proof, together with these exact table computations, recovers the source's affirmative answers to both existence questions.

4 The infinite family and its small dimensions

We retain the two character-theoretic results quoted in the source: Lusztig's criterion for the existence of the cuspidal unipotent character at triangular dimension $n = s(s+1)/2$, and Ohmori's criterion that its real Schur index is two precisely when $\lfloor n/2 \rfloor$ is odd, in odd characteristic. We also retain the quoted Nuzhin generation theorem in its stated range $n \geq 8$.

For completeness about the two smaller triangular dimensions, the subsequent primary classification [2, Theorem 1.1 and following paragraph] says that $\mathrm{PSU}(3, q)$ admits no string C-group representation, whereas $\mathrm{PSU}(6, q)$ admits one of rank three for odd q. These are expressly borrowed classification inputs here. This use does not constitute a recovery of that separate classification paper.

Theorem 3. *Let q be an odd prime power, and let $n = s(s+1)/2$ with $\lfloor n/2 \rfloor$ odd. Then $\mathrm{PSU}(n, q)$ has a string C-group representation and the indicated irreducible character of real Schur index two if $n \geq 6$. When $n = 3$ it has no string C-group representation at any rank.*

Proof. The triangular dimensions below eight are $1, 3, 6$. The parity condition excludes 1. The borrowed classification settles 3 and 6, and the stated Nuzhin theorem supplies rank three for all remaining admissible dimensions. Lusztig and Ohmori give the character in every positive case. There are infinitely many such dimensions: put $s = 8j + 5$, so $n = 32j^2 + 44j + 15 \equiv 3 \pmod 4$. □

The printed Theorem 4 of [1] omits the dimensional qualification, and therefore includes the false $n = 3$ case. The infinite-family conclusion itself survives; the $n = 6$ extension uses the additional input explicitly identified above. The source's final suggestion that these cuspidal characters should always be polytopal is not established by the present argument and remains a separate question in this recovery.

5 Scope of the recovery

The source reports that a particular implementation, choosing one representative from each of 150 conjugacy classes and conjugating s_0 by it, returned 36 markings, one of type $\{5, 5\}$. Those representatives and that run are not specified in the manuscript; the number is an implementation report, not an invariant enumeration of all markings. We neither reproduce that historical run nor promote it to such a classification. Proposition 2 supplies the fully specified witness needed for the principal existence and character conclusions. Its matrices replace the source's unspecified choices. The general fixed-vector formula, column bound, source attributions, corrected parameter range and computational premises are all explicit. No new solution of an open conjecture, formal Lean proof, or compression measurement is claimed by this manuscript.

References

[1] P. J. Cameron, A. Herman and D. Leemans, *String C-groups with real Schur index 2*, arXiv:2201.01313v1 (4 January 2022), six pages; J. Pure Appl. Algebra **226** (2022), 107025. The inspected source is the stated preprint.

[2] D. Leemans and A. Vandenschrick, *On simple groups that admit a string C-group representation*, retained six-page author manuscript, Theorem 1.1 and its following rank-three paragraph. The general classification is used as an input.

[3] The GAP Group and T. Breuer, *GAP Character Table Library*, version 1.3.11, ordinary table `U5(3)`; GAP 4.15.1.